Project Progress - Week 2 of March 2025

Joan Ronquillo

March 10, 2025

This is a file that contains the tracking of the activities related to the OFC project during the second week of March 2025. The activities are divided into groups and a summary of the progress is provided.

1 Initial Status of the Project

The project has made significant progress up until the second week of March 2025. The key achievements include:

- The management of Python modules and compilation of LaTeX projects has been learned.
- A first approach to C++ modular development of the functionalities has been initiated.
- The reading of Draine's paper on radiative corrections in polarizability [1] has been initiated.
- The difficulties in implementing C++ modules have been discussed with Raúl. And intuition on how to proceed has been provided.
- The need to use the Discrete Dipole Approximation (DDA) for the calculation of optical forces for particles of size comparable to the wavelength has been discussed.

So far, the project is developing in some areas:

- The development of the algorithm to obtain the polarizability of the particles.
- The modular development, including makefile documentation and Raul's cpp repository and notes analysis.
- The development of a calculation method for Optical Forces given a polarizability, a field, and a field gradient.

2 Potential Tasks for the Week

The following tasks are proposed for the week of March 10th to March 15th, 2025:

- Exploration of CMakefile, include directory, and modular development (Notion)

- Develop the algorithm to obtain the polarizability of the particles.
- Implement the calculation method for Optical Forces given a polarizability, a field, and a field gradient.
- Documentation of quasiestatic polarizability theory.
- Continue reading Draine's paper on radiative corrections in polarizability [1].
- Analysys of Raúl's `cpp-template-project` repository.
- Check of Raúl's class notes.
- Discuss the progress and difficulties with Raúl.

References

[1] B. T. Draine. "The Discrete-Dipole Approximation and Its Application to Interstellar Graphite Grains". In: *The Astrophysical Journal* 333 (Oct. 1988), p. 848. DOI: `10.1086/166795`.